

Practical No. 5

Aim: Develop a C++ program to create a queue using linked lists and perform enqueue, dequeue, and display operations.

Source-Code:

```
#include <iostream>
using namespace std;
struct Node {
    int data;
    Node* next;
};
Node* front = NULL;
Node* rear = NULL;
void enqueue(int value) {
    Node* newNode = new Node();
    newNode->data = value;
    newNode->next = NULL;

    if (front == NULL) {
        front = rear = newNode;
    } else {
        rear->next = newNode;
        rear = newNode;
    }
    cout << value << " inserted into
queue\n";
}
void dequeue() {
    if (front == NULL) {
        cout << "Queue is empty. Cannot
dequeue.\n";
        return;
    }
    Node* temp = front;
    cout << temp->data << " removed from
queue\n";
    front = front->next;
    delete temp;
}
void display() {
    if (front == NULL) {
        cout << "Queue is empty.\n";
        return;
    }

    Node* temp = front;
    cout << "Queue elements: ";
    while (temp != NULL) {
        cout << temp->data << " ";
```

```
temp = temp->next;
    }
    cout << endl;
}
int main() {
    int choice, value;
    do {
        cout << "\n--- Queue Menu ---\n";
        cout << "1. Enqueue\n";
        cout << "2. Dequeue\n";
        cout << "3. Display\n";
        cout << "4. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Enter value: ";
                cin >> value;
                enqueue(value);
                break;

            case 2:
                dequeue();
                break;

            case 3:
                display();
                break;

            case 4:
                cout << "Exiting program.\n";
                break;

            default:
                cout << "Invalid choice!\n";
        }
    } while (choice != 4);

    return 0;
}
```

Output:

```
--- Queue Menu ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value: 22
22 inserted into queue

--- Queue Menu ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 22

--- Queue Menu ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
22 removed from queue

--- Queue Menu ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 4
Exiting program.
```